

2026-08-22

QC prep lecture 1

Ex 1 (a) What is the inner product between the real vectors $(0, 1, 0, 1)$ and $(0, 1, 1, 1)$?
 (b) What is the inner product between the states $|0101\rangle$ and $|0111\rangle$?

a) $0 \cdot 0 + 1 \cdot 0 + 0 \cdot 1 + 1 \cdot 1 = 1$

b) $\langle 0101 | 0111 \rangle = \langle 5 | 4 \rangle = (00000100) \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{pmatrix} = 0$ (orthogonal)

Ex 2 Compute the result of applying a Hadamard transform to both qubits of $|0\rangle \otimes |1\rangle$ in two ways (the first way using tensor product of vectors, the second using tensor product of matrices), and show that the two results are equal:

$$H|0\rangle \otimes H|1\rangle = (H \otimes H)(|0\rangle \otimes |1\rangle).$$

$$H|0\rangle \otimes H|1\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \otimes \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \end{bmatrix} \otimes \begin{bmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ \frac{1}{2} \\ -\frac{1}{2} \\ \frac{1}{2} \end{bmatrix}$$

$$(H \otimes H)(|0\rangle \otimes |1\rangle) = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \\ -1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 1 \\ -1 \end{bmatrix}$$

Ex 3 Show that a bitflip operation, preceded and followed by Hadamard transforms, equals a phaseflip operation: $HXH = Z$.

$$HXH = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = Z.$$

or

$$HXH|0\rangle = HX|+\rangle = HX\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)\right) = H|+\rangle = |0\rangle = Z|0\rangle$$

$$HXH|1\rangle = HX|-\rangle = HX\left(\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = -H|-\rangle = -|1\rangle = Z|1\rangle$$

and because of linearity this would work for any 2-dim vector, (since $|0\rangle$ & $|1\rangle$ make orthonormal basis)

Ex 4

Show that surrounding a CNOT gate with Hadamard gates switches the role of the control-bit and target-bit of the CNOT: $(H \otimes H) \text{CNOT} (H \otimes H)$ is the 2-qubit gate where the second bit controls whether the first bit is negated (i.e., flipped).

$$\begin{aligned}
 (H \otimes H) \text{CNOT}_{12} (H \otimes H) &= \\
 &= \frac{1}{2} \begin{pmatrix} H & H \\ H & -H \end{pmatrix} \begin{pmatrix} I & 0 \\ 0 & X \end{pmatrix} \begin{pmatrix} H & H \\ H & -H \end{pmatrix} = \\
 &= \frac{1}{2} \begin{pmatrix} H & H \\ H & -H \end{pmatrix} \begin{pmatrix} H & I \\ XH & -XH \end{pmatrix} = \frac{1}{2} \begin{pmatrix} I+Z & I-Z \\ I-Z & I+Z \end{pmatrix} = \\
 \begin{matrix} I+Z = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ I-Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \end{matrix} & \left| \begin{matrix} = \frac{1}{2} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} = \text{CNOT}_{21} \end{matrix} \right.
 \end{aligned}$$

$$\begin{aligned}
 \text{CNOT}_{12} &= \begin{pmatrix} 1 & 0 \\ 0 & X \end{pmatrix} \\
 \text{CNOT}_{21} &= \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & -1 \end{pmatrix} \\
 \because 100 &\rightarrow 100 \\
 101 &\rightarrow 111 \\
 110 &\rightarrow 110 \\
 111 &\rightarrow 101
 \end{aligned}$$

$$\begin{aligned}
 HH &= I \\
 HXH &= Z
 \end{aligned}$$

Ex 5

Simplify the following: $(\langle 0| \otimes I)(\alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle)$.

$$\begin{aligned}
 &= \alpha_{00}|0\rangle + \alpha_{01}|1\rangle \quad \begin{matrix} \xrightarrow{\langle 0|0\rangle = 1} & \xrightarrow{\langle 0|1\rangle = 0} \end{matrix} \quad (\text{unnormalized measurement})
 \end{aligned}$$

Ex 6

Prove that an EPR-pair $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ is an *entangled* state, i.e., that it cannot be written as the tensor product of two separate qubits.

Assume the opposite:

$$\exists \psi, \varphi \text{ s.t.: } \psi \otimes \varphi = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

$$\begin{aligned}
 \psi &= \alpha|0\rangle + \beta|1\rangle \\
 \varphi &= \gamma|0\rangle + \delta|1\rangle
 \end{aligned}
 \left. \vphantom{\begin{aligned} \psi &= \alpha|0\rangle + \beta|1\rangle \\ \varphi &= \gamma|0\rangle + \delta|1\rangle \end{aligned}} \right\} \psi \otimes \varphi = \alpha\gamma|00\rangle + \alpha\delta|01\rangle + \beta\gamma|10\rangle + \beta\delta|11\rangle$$

$$\text{Hence } \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \alpha\gamma \cdot \beta\delta = \alpha\delta \cdot \beta\gamma = 0 \cdot 0$$

contradiction $\therefore \nexists \psi, \varphi$

Ex 7

Suppose we have the state $\frac{1}{\sqrt{2}}(|0\rangle|\phi\rangle + |1\rangle|\psi\rangle)$, where $|\phi\rangle$ and $|\psi\rangle$ are unknown normalized quantum states with the same number of qubits. Suppose we apply a Hadamard gate to the first qubit and then measure that first qubit in the computational basis. Give the probability of measurement outcome 1, as a function of the states $|\phi\rangle$ and $|\psi\rangle$.

$$\begin{aligned}
 (H \otimes I) \left(\frac{1}{\sqrt{2}}(|0\rangle|\phi\rangle + |1\rangle|\psi\rangle) \right) &= \frac{1}{\sqrt{2}}(|+\rangle|\phi\rangle + |-\rangle|\psi\rangle) = \\
 &= \frac{1}{2}(|0\rangle|\phi\rangle + |1\rangle|\phi\rangle + |0\rangle|\psi\rangle - |1\rangle|\psi\rangle) =: |\alpha\rangle
 \end{aligned}$$

$$\text{For } P_1 = |1\rangle\langle 1|, \quad \langle 1|1\rangle = \langle 0|0\rangle = 1, \quad \langle 1|0\rangle = \langle 0|1\rangle = 0$$

(projective measurement $P_1 = |1\rangle\langle 1|$)

$$\begin{aligned}
 P_1 &= \langle \alpha | P_1 | \alpha \rangle = \langle \alpha | 1 \rangle \langle 1 | \alpha \rangle = \frac{1}{4} (\langle \varphi | - \langle \psi |) (| \varphi \rangle - | \psi \rangle) \\
 &= \frac{1}{4} (\underbrace{\langle \varphi | \varphi \rangle}_{\text{normalized}} - \langle \psi | \varphi \rangle - \langle \varphi | \psi \rangle + \underbrace{\langle \psi | \psi \rangle}_{\text{normalized}}) = \\
 &= \frac{1}{4} (2 - \langle \psi | \varphi \rangle - \langle \psi | \varphi \rangle^*) = \frac{1}{2} - \frac{1}{2} \text{Re}(\langle \psi | \varphi \rangle)
 \end{aligned}$$

Ex 8

Give the 2-outcome projective measurement on a 2-qubit space that measures the parity (i.e., sum modulo 2) of 2-bit basis states. Also give the corresponding observable.

$$|a, b\rangle \rightarrow a \oplus b \quad P_0 = |00\rangle\langle 00| + |11\rangle\langle 11|$$

Observable: spectral decomp $P_1 = |01\rangle\langle 01| + |10\rangle\langle 10|$

$$M = \underline{0} \cdot P_0 + \underline{1} \cdot P_1 = |01\rangle\langle 01| + |10\rangle\langle 10| = \begin{bmatrix} 0 & & \\ & 1 & \\ & & 1 \\ & & & 0 \end{bmatrix}$$

outcomes — M eigenvalues, NOT PROBABILITIES

at $|\psi\rangle$,
probability to
measure 0 is
 $\langle \psi | P_0 | \psi \rangle$
which leads to
state
 $\frac{P_0 |\psi\rangle}{\sqrt{\langle \psi | P_0 | \psi \rangle}}$

Ex 9

(H) Show that the success probability of the POVM at the end of Section 1.2.2 can be increased from $1/4$ to $1/(2 + \sqrt{2})$.

Given was meaning $|\psi\rangle \rightarrow$ is $|0\rangle \rightarrow E_0 = \frac{1}{2} |-\rangle\langle -|$
 is $|+\rangle \rightarrow E_1 = \frac{1}{2} |1\rangle\langle 1|$
 not defined $\rightarrow E_2 = I - E_0 - E_1$

Impossible to ideally distinguish
not orthogonal states!

$$\begin{aligned}
 |0\rangle \\
 P_0 &= \langle 0 | \frac{1}{2} |-\rangle\langle -| 0 \rangle = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{4} \\
 P_1 &= \langle 0 | \frac{1}{2} |1\rangle\langle 1| 0 \rangle = 0 \\
 P_2 &= 1 - \frac{1}{4} - 0 = \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 |+\rangle \\
 P_0 &= \langle + | \frac{1}{2} |-\rangle\langle -| + \rangle = 0 \\
 P_1 &= \langle + | \frac{1}{2} |1\rangle\langle 1| + \rangle = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{4} \\
 P_2 &= 1 - 0 - \frac{1}{4} = \frac{3}{4}
 \end{aligned}$$

Cannot make mistake — e.g. $|0\rangle$ and $|-\rangle$ overlap but
 seems no other way to $|+\rangle$ and $|-\rangle$ don't. (same with $|+\rangle$ & $|1\rangle$)
 pick projectors. So let's try adjusting coeffs $\left(\frac{1}{2}\right)$.

$$\langle 0 | C_1 |-\rangle\langle -| 0 \rangle = C_1 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} C_1$$

$$\langle + | C_2 |1\rangle\langle 1| + \rangle = C_2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{2} C_2$$

constraint: $I - C |-\rangle\langle -| - C |1\rangle\langle 1| \geq 0$

So just pick bigger
 C_1 & C_2 (equally).

FC

$$\frac{1}{2} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \begin{pmatrix} 1 & -1 \end{pmatrix}$$

Aim: Find max c s.t. $M := I - C |-\rangle\langle -| - C |1\rangle\langle 1| \geq 0$.

$$M = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \frac{c}{2} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} - c \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 - \frac{c}{2} & \frac{c}{2} \\ \frac{c}{2} & 1 - \frac{3c}{2} \end{pmatrix}$$

$$\left. \begin{aligned} \text{Tr}(M) &= \lambda_1 + \lambda_2 \\ \text{Det}(M) &= \lambda_1 \cdot \lambda_2 \end{aligned} \right\} \text{ if both } \geq 0, \text{ then } M \geq 0$$

Let's try

$$\frac{2}{2}C = \frac{1}{2+\sqrt{2}} \text{ from question.}$$

$$C = \frac{2}{(2+\sqrt{2})(2-\sqrt{2})} = \frac{\dots}{4-2} = 2-\sqrt{2}$$

$$\text{Then: } I - (2-\sqrt{2}) |-\rangle\langle -| - (2-\sqrt{2}) |1\rangle\langle 1| =$$

$$= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{2+\sqrt{2}} \\ \frac{1}{2+\sqrt{2}} & -2 + \frac{3\sqrt{2}}{2} \end{pmatrix} \geq 0 \quad \left[\begin{array}{l} \text{Det} = 0 \quad \rightarrow \text{increasing } c \\ \text{makes det negative.} \\ \text{Trace} = 2\sqrt{2} - 2 > 0 \end{array} \right]$$

Therefore:

$$E_0 = (2-\sqrt{2}) |-\rangle\langle -|$$

$$E_1 = (2-\sqrt{2}) |1\rangle\langle 1|$$

$$E_2 = I - E_1 - E_2$$

$$\begin{aligned} |0\rangle \\ p_0 &= \langle 0 | \frac{1}{2} |-\rangle\langle -| 0\rangle = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{4} \\ p_1 &= \langle 0 | \frac{1}{2} |1\rangle\langle 1| 0\rangle = 0 \\ p_2 &= 1 - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

$$\begin{aligned} |+\rangle \\ p_0 &= \langle + | \frac{1}{2} |-\rangle\langle -| +\rangle = 0 \\ p_1 &= \langle + | \frac{1}{2} |1\rangle\langle 1| +\rangle = \frac{1}{2} \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}} = \frac{1}{4} \\ p_2 &= 1 - \frac{1}{4} = \frac{3}{4} \end{aligned}$$

Ex 10

(H) Prove the quantum no-cloning theorem: there does not exist a 2-qubit unitary U that maps

$$|\phi\rangle|0\rangle \mapsto |\phi\rangle|\phi\rangle$$

for every qubit $|\phi\rangle$.

Unitary preserves inner product ($U^\dagger U = I$, $\langle \psi | \varphi \rangle \rightarrow \langle \psi | U^\dagger U | \varphi \rangle = \langle \psi | \varphi \rangle$)

Assume $\exists U$ s.t. $U|\phi\rangle|0\rangle = |\phi\rangle|\phi\rangle$ and $U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle$. Then

$$\langle 0 | \langle \phi | \psi \rangle | 0 \rangle = \langle \phi | \langle \phi | \psi \rangle | \psi \rangle,$$

$$\langle 0 | 0 \rangle \langle \phi | \psi \rangle = \langle \phi | \psi \rangle \langle \phi | \psi \rangle,$$

$$\langle \phi | \psi \rangle = |\langle \phi | \psi \rangle|^2 \Rightarrow \text{either } |\phi\rangle = |\psi\rangle \text{ or } |\phi\rangle \perp |\psi\rangle. \quad \square$$

Ex 11 Show that unitaries cannot "delete" information: there is no 1-qubit unitary U that maps $|\phi\rangle \mapsto |0\rangle$ for every 1-qubit state $|\phi\rangle$.

Assume $\exists U$ s.t. $U|\phi\rangle = |0\rangle, \forall |\phi\rangle$.

Then also for arbitrary $|\psi\rangle \neq |\phi\rangle$: $U|\psi\rangle = |0\rangle$.

Unitary preserves inner product, thus

$$\langle\phi|\psi\rangle = \langle 0|0\rangle = 1, \text{ which means } |\psi\rangle = |\phi\rangle \text{ (since it's normalized)}$$

Contradiction. $\square$

Ex 12 Suppose Alice and Bob are not entangled. If Alice sends a qubit to Bob, then this can give Bob at most one bit of information about Alice.⁹ However, if they share an EPR-pair, $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, then they can transmit *two* classical bits by sending one qubit over the channel; this is called *superdense coding*. This exercise will show how this works.

(a) They start with a shared EPR-pair, $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$. Alice has classical bits a and b . Suppose she does an X -gate on her half of the EPR-pair if $a = 1$, followed by a Z -gate if $b = 1$ (she does both if $ab = 11$, and neither if $ab = 00$). Write the resulting 2-qubit state for the four different cases that ab could take.

(b) Suppose Alice sends her half of the state to Bob, who now has two qubits. Show that Bob can determine both a and b from his state, using Hadamard and CNOT gates, followed by a measurement in the computational basis.

(a) • $ab = 00$: $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

• $ab = 01$: $(I \otimes Z)\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

• $ab = 10$: $(I \otimes X)\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$

• $ab = 11$: $(I \otimes Z)(I \otimes X)\left(\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)\right) = \frac{1}{\sqrt{2}}(-|01\rangle + |10\rangle)$

(b) • $ab = 00$: $(H \otimes I) \text{CNOT} \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (H \otimes I) \frac{1}{\sqrt{2}}(|00\rangle + |10\rangle) =$
 $= (H \otimes I) |+\rangle |0\rangle = |00\rangle \xrightarrow{\text{measure}} 00 \text{ [label as } ba]$

• $ab = 01$: $(H \otimes I) \text{CNOT} \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle) = (H \otimes I) \frac{1}{\sqrt{2}}(|00\rangle - |10\rangle) =$
 $= (H \otimes I) |-\rangle |0\rangle = |10\rangle \xrightarrow{\text{measure}} 10 \text{ [label as } ba]$

• $ab = 10$: $(H \otimes I) \text{CNOT} \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle) = (H \otimes I) \frac{1}{\sqrt{2}}(|01\rangle + |11\rangle) =$
 $= (H \otimes I) |+\rangle |1\rangle = |01\rangle \xrightarrow{\text{measure}} 01 \text{ [label as } ba]$

• $ab = 11$: $(H \otimes I) \text{CNOT} \frac{1}{\sqrt{2}}(-|01\rangle + |10\rangle) = (H \otimes I) \frac{1}{\sqrt{2}}(-|01\rangle + |11\rangle) =$
 $= (H \otimes I) |-\rangle |1\rangle = -|11\rangle \xrightarrow{\text{measure}} 11 \text{ [label as } ba]$

Ex 13 Alice and Bob share an EPR-pair, $|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$.

- (a) Let C be a 2×2 matrix. Show that $\text{Tr}((C \otimes I)|\psi\rangle\langle\psi|) = \frac{1}{2}\text{Tr}(C)$.
 (b) (H) Alice could apply one of the 4 Pauli matrices (I, X, Y, Z) to her qubit. Use part (a) to show that the 4 resulting 2-qubit states form an orthonormal set.
 (c) Suppose Alice applies one of the 4 Pauli matrices to her qubit and then sends that qubit to Bob. Give the 4 projectors of a 4-outcome projective measurement that Bob could do on his 2 qubits to find out which Pauli matrix Alice actually applied.

(a) $\text{Tr}(A) = \sum_w \langle w|A|w\rangle = \sum_i \lambda_i$

$\text{Tr}((C \otimes I)|\psi\rangle\langle\psi|) \stackrel{\text{cyclic}}{=} \text{Tr}(|\psi\rangle\langle\psi|(C \otimes I)) =$
 $\stackrel{\text{partial trace property}}{=} \text{Tr}(\text{Tr}_B(|\psi\rangle\langle\psi|)C) \stackrel{\text{EPR partial trace is } \frac{I}{2}}{=} \text{Tr}(\frac{I}{2} \cdot C) \stackrel{\text{Tr}(aA) = a\text{Tr}(A)}{=} \frac{1}{2}\text{Tr}(C)$
or just expanding and calculating manually

(b) $I: (I \otimes I)|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$

$X: (X \otimes I)|\psi\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle)$

$Y: (Y \otimes I)|\psi\rangle = \frac{1}{\sqrt{2}}(i|10\rangle - i|01\rangle) = \frac{i}{\sqrt{2}}(|10\rangle - |01\rangle)$

$Z: (Z \otimes I)|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)$

for any pair $P, Q \in \{I, X, Y, Z\}$ we need to have $\langle\psi_P|\psi_Q\rangle = 0$.

or just calculating each pair to be
 $\langle\varphi_1|\varphi_2\rangle = 0$
since ψ is pure

$\langle\psi_P|\psi_Q\rangle = \langle\psi|(P \otimes I)^\dagger(Q \otimes I)|\psi\rangle \stackrel{\text{since } \psi \text{ is pure}}{=} \text{Tr}((P^\dagger Q \otimes I)|\psi\rangle\langle\psi|) =$
 $= \frac{1}{2}\text{Tr}(P^\dagger Q) \stackrel{\text{Pauli Hermitian}}{=} \frac{1}{2}\text{Tr}(PQ).$

if $P=Q$: $\frac{1}{2}\text{Tr}(P^\dagger P) \stackrel{\text{Pauli Hermitian}}{=} \frac{1}{2}\text{Tr}(P^2) \stackrel{\text{Pauli}^2=I}{=} \frac{1}{2}\text{Tr}(I) = \frac{1}{2} \cdot 2 = 1$

if $P \neq Q$: if $P=I$ or $Q=I$: $\text{Tr}(P) \stackrel{\text{Pauli traceless}}{=} 0$. NORMAL

else: $\text{tr}(XY) = \text{tr}(IZ) = 0$

$\text{tr}(XZ) = \text{tr}(ZX) = \text{tr}(iY) = 0$

$\text{tr}(YZ) = \text{tr}(iX) = 0$

$ZX = iY$

$XY = iZ$

$YZ = iX$

$X \rightarrow Y$
 $Y \rightarrow Z$
 $Z \rightarrow X$

□

$$\begin{aligned}
 (c) \quad P_I &= |\psi\rangle\langle\psi|, |\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \\
 P_X &= |\psi_x\rangle\langle\psi_x|, |\psi_x\rangle = \frac{1}{\sqrt{2}}(|10\rangle + |01\rangle) \\
 P_Y &= |\psi_y\rangle\langle\psi_y|, |\psi_y\rangle = \frac{1}{\sqrt{2}}(i|10\rangle - i|01\rangle) \\
 P_Z &= |\psi_z\rangle\langle\psi_z|, |\psi_z\rangle = \frac{1}{\sqrt{2}}(|00\rangle - |11\rangle)
 \end{aligned}$$

Need to have

$$\langle\psi|P_I|\psi\rangle = 1$$

$$\langle\psi_x|P_X|\psi_x\rangle = 1$$

etc.

Possible since
orthonormal.

Ex 14

$$\text{Let } \theta \in [0, 2\pi), U_\theta = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}, |\phi\rangle = U_\theta|0\rangle \text{ and } |\phi^\perp\rangle = U_\theta|1\rangle.$$

- Show that $ZX|\phi^\perp\rangle = |\phi\rangle$.
- Show that an EPR-pair, $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, can also be written as $\frac{1}{\sqrt{2}}(|\phi\rangle|\phi\rangle + |\phi^\perp\rangle|\phi^\perp\rangle)$.
- Suppose Alice and Bob start with an EPR-pair. Alice applies U_θ^{-1} to her qubit and then measures it in the computational basis. What pure state does Bob have if her outcome was 0, and what pure state does he have if her outcome was 1?
- Suppose Alice knows the number θ but Bob does not. Give a protocol that uses one EPR-pair and 1 classical bit of communication where Bob ends up with the qubit $|\phi\rangle$ (in contrast to general teleportation of an unknown qubit, which uses 1 EPR-pair and 2 bits of communication).

$$\begin{aligned}
 (a) \quad ZX|\phi^\perp\rangle &= ZX(-\sin\theta|0\rangle + \cos\theta|1\rangle) = Z(-\sin\theta|1\rangle + \cos\theta|0\rangle) = \\
 &= \sin\theta|1\rangle + \cos\theta|0\rangle = |\phi\rangle
 \end{aligned}$$

$$\begin{aligned}
 (b) \quad \frac{1}{\sqrt{2}}(|\phi\rangle|\phi\rangle + |\phi^\perp\rangle|\phi^\perp\rangle) &= \frac{1}{\sqrt{2}}(U_\theta \otimes U_\theta |00\rangle + U_\theta \otimes U_\theta |11\rangle) = \\
 &= (U_\theta \otimes U_\theta) \cdot \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (I \otimes U_\theta)(U_\theta \otimes I)|\psi\rangle = \\
 &\stackrel{\text{transpose trick for max entangled state}}{=} (I \otimes U_\theta)(I \otimes U_\theta^\dagger)|\psi\rangle = (I \otimes U_\theta U_\theta^\dagger)|\psi\rangle = |\psi\rangle
 \end{aligned}$$

$\hookrightarrow = I$ since U_θ

$$\begin{aligned}
 \rightarrow \langle\phi|\phi^\perp\rangle &= (\langle 0|\cos\theta + \langle 1|\sin\theta)(\cos\theta|1\rangle - \sin\theta|0\rangle) = \\
 &= -\cos\theta\sin\theta + \sin\theta\cos\theta = 0
 \end{aligned}$$

$$\sin^2\theta + \cos^2\theta = 1$$

$$(c) \quad U_\theta^{-1} = U_\theta^\dagger = \begin{pmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix}$$

$$(U_\theta^\dagger \otimes I) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) = (I \otimes U_\theta) \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) =$$

$$= \frac{1}{\sqrt{2}} (|0\rangle|\varphi\rangle + |1\rangle|\varphi^\perp\rangle)$$

$\xrightarrow{A \rightarrow 1}$ Bob: $|\varphi^\perp\rangle$

$\xrightarrow{A \rightarrow 0}$ Bob: $|\varphi\rangle$

(d) Do the (c) part process first. Then Alice sends Bob her measurement result. One classical bit will tell Bob to switch between $|\varphi\rangle$ and $|\varphi^\perp\rangle$ or not.

If Bob gets 0 $\rightarrow$ doesn't change the state has $|\varphi\rangle$

If Bob gets 1 $\rightarrow$ applies ZX to the qubit has $|\varphi^\perp\rangle$
 $\rightarrow ZX|\varphi^\perp\rangle = |\varphi\rangle$
 (see (a))